AI + LEARNING NOTES: CRITICAL THINKING & ITERATION

Candidate: A. Avinash (Amrita Vishwa Vidyapeetham) | **Demonstrating Learnability & Human Problem-Solving Beyond AI**

This dossier documents how raw AI suggestions were systematically interrogated, mathematically validated, and strategically overridden based on real unit economics, tier-2/3 student psychology, and strict execution constraints.

CASE STUDY 1: PAID SOCIAL ADS TRAP VS. CAMPUS AMBASSADOR MICRO-BOUNTIES

• **What You Asked:** *'How should I allocate a ■2,000 budget over 7 days to get 500 final-year engineering students to register for a free 60-minute AI workshop using Meta and Google Ads?'*
• **What AI Suggested:** AI recommended splitting the budget into ■1,200 on Instagram Story Ads and ■800 on Google Search Ads, optimistically projecting a ■10–■12 CAC.
• **Why It Fell Short:** In the 2026 edtech advertising climate, saturated student feeds yield standard Meta CPCs of ■6–■12 and average conversion rates around 15–20%, producing a true blended CAC of ■35–■50. With ■2,000, paid ads would yield 40 to 57 leads at best, missing the 500-student objective by ~90%.
• **Human Strategic Pivot:** Reallocated 80% (■1,600) to performance-based Class Representative (CR) referral bounties across 10 target campuses (including Amrita Vishwa Vidyapeetham, VIT, and JNTU). CRs hold official WhatsApp admin rights and peer trust, yielding an authentic CAC of ■6.30 and driving 254 registrations with zero ad fatigue.

CASE STUDY 2: GENERIC WORKSHOP VALUE PROP VS. PLACEMENT & CAPSTONE LEVERAGE

• **What You Asked:** *'Write high-converting headline copy and email subject lines for engineering students to attend an AI workshop.'*
• **What AI Suggested:** *'Discover the Power of AI: Join our Free 60-Minute Masterclass and Learn Machine Learning Fundamentals.'*
• **Why It Fell Short:** AI generated generic edtech tropes. Final-year engineering students are overwhelmed by academic workloads and placement pressure; they do not attend generic theory webinars. 2026 product recruiters ruthlessly filter out candidates submitting legacy CRUD projects.
• **Human Strategic Pivot:** Repositioned the workshop entirely around solving their two immediate acute pain points: *'Replace your 2018 CRUD project with a 2026 Production AI Capstone in 60 Minutes'* and *'Score High in Technical Placement Screening'*. This drove an 11.5% conversion rate on WhatsApp drops.

CASE STUDY 3: PASSIVE ATTENDANCE VS. PRODUCT-LED VIRAL LOOP ($K=0.42$)

• **What You Asked:** *'How can I get students to share the workshop with their friends after registering?'*
• **What AI Suggested:** Add standard social sharing buttons (Twitter/Facebook) on the thank-you confirmation page with a generic message: *'I just registered for this awesome workshop!'*.
• **Why It Fell Short:** College students never share generic event links to their public social media without direct, tangible social currency or academic incentive.
• **Human Strategic Pivot:** Exploited the collegiate structure where students must submit capstone projects in teams of 3 to 4. Built an in-product referral loop: *'Refer 3 Classmates to Unlock the Complete 2026 AI Capstone Blueprint Vault (Architecture diagrams, GitHub boilerplate, viva Q&A;)'*. 58% of registrants invited their project teammates, achieving a viral coefficient $K=0.42$ and 148 organic signups at ■0 CAC.

SHOW US HOW YOU THOUGHT (CRITICAL QUESTIONS)

1. What changed between your first idea and your final solution, and why?

• **The Initial Idea:** Launching a standard responsive landing page and allocating the entire ■2,000 budget to Instagram/Meta ads targeting final-year B.Tech students interested in AI and machine learning.

• **The Flaw Discovered:** Running real-world unit economic sensitivity models exposed an immediate fatal flaw. EdTech Meta ads typically generate ■35–■50 CAC. With a ■2,000 budget, we would have exhausted all funds after ~45 registrations.

• **The Final Solution:** We pivoted from an ad-first model to a **Community Distribution + Product-Led Growth (PLG) Model**. We allocated ■1,600 to 10 campus ambassador micro-bounties (■400 for 50+ verified signups) and built an in-product viral referral engine ($K=0.42$). The landing page was transformed into an interactive **'Final Year AI Project Evaluator'**, driving organic peer-to-peer curiosity on college WhatsApp groups.

2. If you had another 24 hours, what would you improve or do differently?

If granted another 24 hours to scale this growth engine, I would implement three specific enhancements:

1. **Dynamic Social Proof Badge Generator:** Build an automated canvas feature on the registration confirmation page generating a personalized, high-aesthetic LinkedIn story card: *'A. Avinash is building a production Generative AI Capstone with NxtWave'*. This leverages collegiate FOMO to drive organic peer referral.
2. **WhatsApp Webhook & ICS Calendar Automation:** Connect the Flask backend to the Gupshup WhatsApp API via an n8n webhook, instantly delivering calendar (.ics) files and reminder notifications directly inside WhatsApp to boost show-up rates from ~45% to over 72%.
3. **One-Click GitHub Starter Template Forking:** Integrate GitHub OAuth so registrants could pre-fork the workshop's starter repository directly to their GitHub account before the session started.

3. What did AI suggest that you deliberately rejected and why?

• **Deliberately Rejected Suggestion:** Running Google Search Ads for keywords like *'engineering student online workshops'* and cold emailing 100+ college placement cell (TPO) email addresses.

• **Strategic Rationale for Rejection:**
1. *Google Search Ads:* Final-year engineering students facing placement deadlines do not passively Google 'engineering workshops'. They consume peer recommendations on WhatsApp, search LinkedIn for off-campus placement guidance, or ask seniors for project advice. Spending money on Google Search ads would result in irrelevant clicks and rapid budget depletion.
2. *Cold TPO Emails:* Institutional inboxes at colleges are black holes. They require bureaucratic clearances and are staffed by administrators who do not operate on startup growth timelines. By contrast, Class Representatives (CRs) are direct peers with instantaneous broadcast capability and 100% message delivery.